Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694 · 089 632 5829 · LINE solaress1
www.solaress.org · www.โซลาร์เอสส์.com

SOLAR ESS CARE · PROJECT RATIONALE

Solar for Everyone.

Affordable solar for every household — built from tested equipment in our own stock, installed and cared for by Solar ESS.

SAFETY FIRST | TESTED & AUDITED | THERMAL SCAN AFTER INSTALL | SCHEDULED MONITORING | 5-YEAR SOLAR ESS WARRANTY | MUCH LOWER PRICE

30 October 2026 · Rev 01

Solar ESS Care is a Solar ESS programme that puts verified, unused or still-serviceable equipment back to work, so that a complete, safe solar system costs much less.

Solar should not only be for those who can afford it.



Together with our affiliated network, Solar ESS has equipment in stock that has either never been used, or has been used and is still in good working condition: lithium (LiFePO4) battery cells, breakers, surge protectors, combiner boxes and AC/DC cables. It is new or end-of-line stock, often without a big-brand label on the finished system, and it still performs very well.

Without the right knowledge, this material gets thrown away: old looks worn out and unsafe, while new feels safer, better and more relaxing. At Solar ESS Care we would rather use materials until the end of their life, to cut garbage and waste for the world, and share these resources to produce clean energy. But we must verify every item, label it as safe, and stay conscious of the potential risks for everyone.

Solar ESS Care is about making the most of these resources and putting them back to work. The equipment may not be brand new or the latest model, and it will not meet every customer's expectations. But it is not junk or waste: it is usable and it is safe. Every item is checked and verified before it goes into a system, and we keep improving how we test and verify it.

Solar for Everyone puts this equipment to work. We build complete, safe systems from it and offer them at a much lower price, so households, small farms, schools and community buildings that could never afford a standard system can cut their electricity bills too.

THE RATIONALE

Use it with knowledge, until the end.

Electrical and electronic equipment is rarely used to the end of its working life. Most of it is thrown away while it still works, because of fear and because we do not know it well enough. The evidence is clear.

62 million t
e-waste produced worldwide in 2022, up 82% since 2010

22.3%
of it properly collected and recycled

US$62 billion
of recoverable materials lost every year

439,000 t
of e-waste a year in Thailand

01

Thrown away by fear, not by failure

With no clear way to mark the line between safe and unsafe, materials get thrown away too easily. Often we know a part can still be used, but something new makes us feel safer and more relaxed.

02

Age is not what makes solar unsafe

Studies of solar fires by Fraunhofer ISE and TÜV Rheinland found the main causes were installation faults, above all badly made DC connectors, and that fires typically occur in new systems. Workmanship and checks matter more than the age of a part.

03

Solar parts can be measured

Battery cells can be tested for capacity and internal resistance, breakers for trip function and cables for insulation resistance. So we keep or reject by measurement, with calibrated instruments, not by age or looks, and we monitor every system online.

04

Lower cost, more people with solar

Using verified stock for parts like combiners, cables and conduit lowers the price of a complete system, so households, small farms, schools and community buildings can afford solar and cut their bills too.

We would rather use materials with knowledge until the end than throw them away. You are the customer, and of course the choice is yours.

SOLAR ESS www.solaress.org

HOW WE KEEP IT SAFE

The same standard as every Solar ESS system.

01

Tested first

Every battery cell, breaker, surge protector and cable is checked for capacity, condition and safety before use.

02

Audited design

Our engineers design and check every system, with the right breaker, fuse and cable size for each circuit.

03

Thermal scan

After installation we thermal-scan every connection, breaker and string to catch hot spots before we hand over.

04

5-year warranty

Solar ESS covers installation and parts for 5 years, with the same workmanship and after-sales care as every system.

05

Monitoring

Once connected to the inverter, your system is monitored online, so Solar ESS can spot problems early.

WE USE IT OURSELVES

The same equipment is in our own homes.

Most customers want a new roll of cable and a new set of breakers, and we understand that. But at our own houses, we, the installers, run our solar systems on second-hand breakers and used cables. Why? Because these materials are still usable.

So we make solar more affordable, yet safe and reliable.

Is it safe?

Yes. We test the insulation of every cable, test the magnetic trip of every breaker and audit every part of the system before we use it, at our homes and at yours.

Will it last?

Honestly, some parts may not last the full five years of the warranty. When that happens, we analyse the cause and fix or replace the part. That is the responsibility we take on.

Will it cause a fire?

Most fires in solar systems start from poor installation: loose or badly made DC connections, mismatched connectors, and cables or protection of the wrong size. They are rarely caused by a part simply being second-hand. That is why we check every connection with a thermal scan after installation.

SOLAR ESS www.solaress.org

WHO IT IS FOR

Households and communities on a tight budget.

Families on a tight budget

Lights, fans, a fridge and phone charging powered by the sun.

Small farms

Daytime power for pumps, sheds and equipment.

Schools, temples & community halls

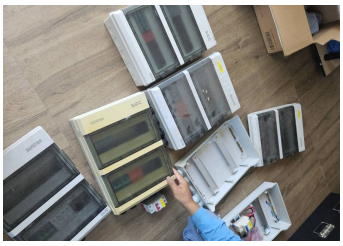
Lower bills for the buildings a community shares.

Remote homes

Small battery systems where the grid is weak or far away.

FROM OUR STOCK



We will be honest with you. These systems will not look as polished as a new branded installation: boxes may be older models, labels may not match and some parts are end-of-line. We accept that, because safety is our first priority, not looks. Every part is tested, every design is audited and every installation is thermal-scanned. We know some parts will need to be fixed or replaced at some point, and we take that responsibility: Solar ESS gives a **5-year warranty** on every Solar for Everyone system.

JOIN THE PROGRAMME

Tell us what you need to power.

Send us a photo of your electricity bill and roof, and we will suggest a Solar for Everyone system that fits your budget.

Phone 063 151 6694 · 089 632 5829 · LINE solaress1 · www.solaress.org · www.โซลาร์เอสส์.com

Sources: UN Global E-waste Monitor 2024 (ITU / UNITAR) (https://ewastemonitor.info/the-global-e-waste-monitor-2024/) · UNITAR press release, 20 March 2024 (https://www.unitar.org/about/news-stories/press/global-e-waste-monitor-2024-electronic-waste-rising-five-times-faster-documented-e-waste-recycling) · The Nation Thailand, September 2025 (https://www.nationthailand.com/sustaination/40055801) · Fraunhofer ISE / TÜV Rheinland, PV fire risk study (Laukamp et al., 28th EU PVSEC 2013) (https://www.ise.fraunhofer.de/de/veroeffentlichungen/konferenzbeitraege/konferenzbeitraege-2013/28th-eupvsec/laukamp.pdf) · Solar Builder, “Rooftop solar fires are on the rise” (2020) (https://solarbuildermag.com/news/pull-the-alarm-rooftop-solar-fires-are-on-the-rise-studies-point-to-two-possible-causes/)